

Cloud Computing Fundamentals

1. Overview

During my Business Computing module, I studied the fundamentals of cloud computing and its relationship with modern data processing and analytics.

Cloud computing involves delivering applications and computing services over the internet. I learned that cloud computing can provide access to software, infrastructure and development platforms without an organisation necessarily having to own and maintain all of the underlying physical computing infrastructure itself.

This evidence summarises my understanding of the cloud computing concepts covered during the module and how they relate to my wider studies in FinTech and data analytics.

2. Cloud Service Models

I studied three important ways in which computing services can be delivered through the cloud:

Software as a Service (SaaS)

Software applications are delivered as a service over the internet, allowing users to access software without having to manage the underlying infrastructure.

Infrastructure as a Service (IaaS)

Computing infrastructure is provided as a service. An example discussed during the module was Amazon EC2.

Platform as a Service (PaaS)

A cloud platform is provided for developing and running applications. Examples discussed during the module included Google AppEngine and Microsoft Azure.

These service models demonstrate that cloud computing can provide different levels of computing resources depending on the requirements of an organisation.

3. Utility Computing and Cloud Economics

Another concept I studied was **utility computing**, where computing resources can be provided using a pay-as-you-go model.

Rather than purchasing enough physical computing infrastructure to meet the highest possible level of demand, cloud users can use resources according to their requirements.

This can provide benefits such as:

- Reduced upfront infrastructure costs.
- Pay-as-you-go billing.
- Access to computing resources when required.
- Reduced risk of maintaining large amounts of unused computing capacity.

I also learned that incorrectly estimating computing requirements can create problems.

Over-provisioning means providing significantly more computing capacity than required, potentially resulting in unused resources and unnecessary cost.

Under-provisioning means providing insufficient capacity to meet demand, which could result in lost users, reduced service quality or lost revenue.

Cloud computing can help organisations respond more flexibly to changing demand.

4. Why Cloud Computing Has Become Important

The development of cloud computing has been supported by both technological and business factors.

Technological developments include widespread broadband internet access, improvements in virtualisation technology and experience operating very large data centres.

From a business perspective, cloud computing can reduce the requirement for large initial capital expenditure and provide a pay-as-you-go approach to computing resources.

Large technology organisations have also contributed to the growth of cloud computing and large-scale computing infrastructure.

5. Cloud Computing and the Data-Driven World

Cloud computing is particularly relevant to data analytics because organisations increasingly generate and analyse very large amounts of data.

Examples of data discussed during my studies included:

- Corporate sales data.
- Stock market transactions.
- Scientific databases.
- Transportation data.
- Social interaction data.
- Medical and patient records.

Within business, data can be analysed to help managers and analysts understand organisational performance and support decision-making.

This is particularly relevant to my FinTech and Data Analytics studies because financial organisations generate large amounts of transactional and market-related information.

6. Data Warehousing and Analytics

I also studied the relationship between data warehousing, data analytics and decision-support systems.

Data warehouses can contain historical or point-in-time transactional information and can be designed primarily for querying and analysis rather than frequent updating.

Managers and analysts can use these systems to investigate business information and support decision-making.

As the amount of data generated by businesses and web-based systems increases, scalable technologies become increasingly important for processing and analysing that information.

7. Large-Scale Data Processing and MapReduce

As part of the cloud computing topic, I was introduced to **MapReduce** as a model for processing large datasets using distributed computing resources.

My understanding of the basic process is:

Map: Data is processed to extract relevant information.

Reduce: Results produced by the Map stage are aggregated.

The underlying computing system can distribute processing across multiple machines. This makes the approach relevant when datasets become too large or computationally demanding for straightforward processing on a single machine.

8. Hadoop, Hive and HiveQL

I was also introduced to **Hadoop** and **Hive** in the context of large-scale data analysis.

Hive provides data warehousing capabilities based on Hadoop and allows data stored within a cluster to be queried using **HiveQL (HQL)**, which has similarities to SQL.

Examples of operations available through HiveQL include:

- SELECT to retrieve columns.
- WHERE to filter rows.
- JOIN to combine tables.
- GROUP BY to perform aggregations.
- CREATE, DROP and ALTER for managing tables.

This was particularly relevant to my development in SQL because it demonstrated how familiar database concepts can also be applied when working with much larger datasets.

9. Connection to My Practical Data Analytics Skills

The accompanying practical work also developed my understanding of combining datasets using **Pandas**.

I studied `pd.merge()` and how relational data can be combined using different types of joins, including:

- One-to-one joins.
- Many-to-one joins.
- Many-to-many joins.
- Inner joins.
- Outer joins.
- Left and right joins.

This helped me understand the relationship between database concepts and practical data analysis in Python.

These concepts complement my wider development in **Python, Pandas and SQL** and demonstrate how data from different sources can be combined and analysed.

10. Application to Financial Data

These concepts can be applied to a financial data scenario.

For example, a financial organisation could generate large volumes of transaction and market data. Cloud computing could provide computing resources to support the storage and processing of this information.

Data-processing technologies could then be used to transform or aggregate large datasets, while database and analytical tools could allow analysts to query the resulting information.

The results could ultimately support activities such as financial analysis, reporting and business decision-making.

This demonstrates the relationship between the cloud computing concepts I studied and my broader FinTech and Data Analytics degree.

11. Conclusion

Through the Business Computing module, I developed a foundational understanding of cloud computing, including SaaS, IaaS, PaaS, utility computing and pay-as-you-go computing.

I also studied the economic reasons for cloud adoption, the challenges of over-provisioning and under-provisioning, and the importance of scalable computing for increasingly data-driven organisations.

The module introduced me to large-scale data-processing concepts including MapReduce, Hadoop, Hive and HiveQL and helped me understand how these technologies relate to databases, data warehousing and analytics.

Alongside practical work involving Python, Pandas and dataset merging, this provided me with a foundation for understanding how modern computing technologies can support large-scale business and financial data analysis.